

ZED RAFAEL SARANILLO

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EDUCATION

San Jose State University

San Jose, CA

Bachelors of Science in Software Engineering

Expected May 2028

Relevant Coursework: Data Structures and Algorithms, Assembly Language, Engineering Statistics, Software Engineering I

Mission College

Santa Clara, CA

Associate of Arts in Math

Graduated May 2026

Relevant Coursework: Linear Algebra, Discrete Math, Computing for Engineers

PROJECTS

TriGro | *React, FastAPI, Docker, PostgreSQL, REST APIs*

Sep 2026 - Present

Personal Project

- Developing full-stack inventory application with React, FastAPI, and PostgreSQL for persistent item tracking
- Implementing **CRUD** flow enabling users to add/manage inventory items from frontend to database
- Developing backend schema and API endpoints to support inventory search functionality

Smiley Financial Bot | *Python, IBM Cloud, Git/Github*

Apr 2026 - May 2026

Watsonx Hackathon

- Engineered Python backend logic to orchestrate **5+ agent workflows**, supporting multiple-user requests
- Integrated IBM watsonx.ai to power context-aware financial analysis across **7 specialized agents**
- Designed routing logic to coordinate task delegation and communication across all 7 agents

Scalegrim | *Asynchronous Programming, Design Patterns, C#, Unity*

Jun 2026 - Present

Personal Project

- Developing card-based deckbuilder using OOP principles across 20+ interconnected gameplay systems
- Implementing data-driven card mechanics with 15+ ScriptableObjects, **supporting 40+ unique cards**
- Designing event-driven architecture handling combat, turn order, and card interactions across 10+ systems
- Build asynchronous gameplay sequencing with C#'s Task/async/await, coordinating 10+ multi-step actions

Obstacle Avoiding Car | *Arduino, C++, Embedded Systems*

Apr 2025 - May 2025

Solo Final Project at Mission College

- Engineered autonomous obstacle-avoidance vehicle, integrating **3 core hardware components**
- Implemented a finite state machine across **4 navigation states** for reversing, scanning, and turning
- Developed 2 control modes with joystick input and debounced button switching
- Integrated real-time distance sensing with **2 feedback systems** using LEDs and piezo

Disoriented driving | *UI/UX, CSS, Unity Version Control*

Jun 2026

Game Jam

- Built responsive UI systems in Unity using CSS, supporting 5+ interactive gameplay elements
- Connected UI to gameplay through event-driven systems, enabling 5+ real-time interactions
- Developed 2 control modes with joystick input and debounced button switching
- Engineered reusable collectible interactions using OOP, supporting 5+ collectible behaviors

TECHNICAL SKILLS

Languages: C#, Java, Python, C++, HTML/CSS

Frameworks/Libraries: React.js, Node.js, FastAPI, Unity

Developer Tools: Git/GitHub, VS Code, Visual Studio, Unity Version Control, IBM Cloud

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Game Design Patterns, Event-Driven Architecture, Asynchronous Programming